Домашняя работа по установке и настройке Suricata

Введение

Цель работы: освоить установку и базовую настройку IDS Suricata в виртуальной среде Linux, а также получить практические навыки создания пользовательских сигнатур для обнаружения сетевых атак и анализа логов срабатывания.

Среда выполнения: Ubuntu (виртуальная машина VirtualBox/VMware).
Важное условие тестирования: Ввиду особенностей виртуализации (сеть NAT), трафик, идущий на внешний IP-адрес, может не проходить через физический драйвер виртуальной сетевой карты (`enp0s3`). Поэтому для наглядного и стабильного тестирования сигнатур Suricata будет настроена на прослушивание интерфейса `lo` (loopback), а тестовые запросы будут выполняться локально.

1. Установка и проверка Suricata

bash

```
sudo apt update
sudo apt install -y software-properties-common
sudo add-apt-repository ppa:oisf/suricata-stable -y
sudo apt update
```

Рекомендуется сразу установить `jq` для удобного просмотра JSON-логов .

bash

```
sudo apt install -y suricata jq
```

Проверка версии и статуса

bash

```
suricata --build-info
```

```
admen@ubuntu:~/work/suricataHW$ suricata --build-info
sudo systemctl status suricata
This is Suricata version 8.0.7 RELEASE
Features: NFQ PCAP_SET_BUFF AF_PACKET HAVE_PACKET_FANOUT LIBCAP_NG LIBNET1.1 HAVE_HTP_URI_NORMALIZE_HOOK PCRE_JIT HAVE_NSS HTTP2_DECOMPRESSION HAVE_LUA HAVE_JA3 HAVE_JA4 HAVE_LIBJANSSO
N TLS TLS_C11 MAGIC RUST POPCNT64
SIMD support: SSE_2
Atomic intrinsics: 1 2 4 8 byte(s)
64-bits, Little-endian architecture
GCC version 13.3.0, C version 201112
compiled with _FORTIFY_SOURCE=2
L1 cache line size (CLS)=64
thread local storage method: _Thread_local
compiled with LibHTP v8.0.7

Suricata Configuration:
  AF_PACKET support:                       yes
  AF_XDP support:                          no
  DPDK support:                            no
  eBPF support:                            no
  XDP support:                             no
  PF_RING support:                         no
  NFQueue support:                         yes
  NFLOG support:                           no
  IPFW support:                            no
  Netmap support:                          no
  DAG enabled:                             no
  Napatech enabled:                        no
  WinDivert enabled:                       no
  Npcap support:

  Unix socket enabled:                     yes
  Detection enabled:                       yes

  Libmagic support:                        yes
  libjansson support:                      yes
  hiredis support:                         yes
  hiredis async with libevent:             yes
  PCRE jit:                                yes
  GeoIP2 support:                          yes
  JA3 support:                             yes
  JA4 support:                             yes
  Hyperscan support:                       yes
  Hwloc support:                           no
  Libnet support:                          yes
  ---
  Host:                                    x86_64-pc-linux-gnu
  Compiler:                                gcc (exec name) / g++ (real)
  GCC Protect enabled:                     yes
  GCC march native enabled:                no
  GCC Profile enabled:                     no
  Position Independent Executable enabled: yes
  CFLAGS                                   -g -O2 -fno-omit-frame-pointer -mno-omit-leaf-frame-pointer -ffile-prefix-map=/build/suricata-tUjEIC/suricata-8.0.7=. -fstack-protector-stron
g -fstack-clash-protection -Wformat -Werror=format-security -fcf-protection -fdebug-prefix-map=/build/suricata-tUjEIC/suricata-8.0.7=/usr/src/suricata-1:8.0.7-0ubuntu0 -fPIC -DOS_LINUX
 -std=c11
  PCAP_CFLAGS                              -I/usr/include/dbus-1.0 -I/usr/lib/x86_64-linux-gnu/dbus-1.0/include -I/usr/include/libnl3
  SECCFLAGS                                -fstack-protector -D_FORTIFY_SOURCE=2 -Wformat -Wformat-security
× suricata.service - Suricata IDS/IPS/NSM/FW daemon
     Loaded: loaded (/usr/lib/systemd/system/suricata.service; enabled; preset: enabled)
     Active: failed (Result: exit-code) since Sun 2026-10-25 16:39:58 UTC; 58min ago
   Duration: 163ms
       Docs: man:suricata(8)
             man:suricatasc(8)
             https://suricata.io/documentation/
    Process: 10052 ExecStartPre=/bin/rm -f /run/suricata.pid (code=exited, status=0/SUCCESS)
    Process: 10054 ExecStart=/usr/bin/suricata --af-packet -c /etc/suricata/suricata.yaml --pidfile /run/suricata.pid --user suricata --group suricata (code=exited, status=1/FAILURE)
   Main PID: 10054 (code=exited, status=1/FAILURE)
        CPU: 114ms

окт 25 16:39:58 ubuntu systemd[1]: suricata.service: Scheduled restart job, restart counter is at 5.
окт 25 16:39:58 ubuntu systemd[1]: suricata.service: Start request repeated too quickly.
окт 25 16:39:58 ubuntu systemd[1]: suricata.service: Failed with result 'exit-code'.
окт 25 16:39:58 ubuntu systemd[1]: Failed to start suricata.service - Suricata IDS/IPS/NSM/FW daemon.
admen@ubuntu:~/work/suricataHW$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:6a:f7:b7 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
       valid_lft 81256sec preferred_lft 81256sec
    inet6 fd17:625c:f037:2:b538:18e9:9abe:913/64 scope global temporary dynamic
       valid_lft 86256sec preferred_lft 14256sec
    inet6 fd17:625c:f037:2:a00:27ff:fe6a:f7b7/64 scope global dynamic mngtmpaddr
       valid_lft 86256sec preferred_lft 14256sec
    inet6 fe80::a00:27ff:fe6a:f7b7/64 scope link
       valid_lft forever preferred_lft forever
admen@ubuntu:~/work/suricataHW$
```

2. Настройка `suricata.yaml`

Откройте конфигурационный файл:

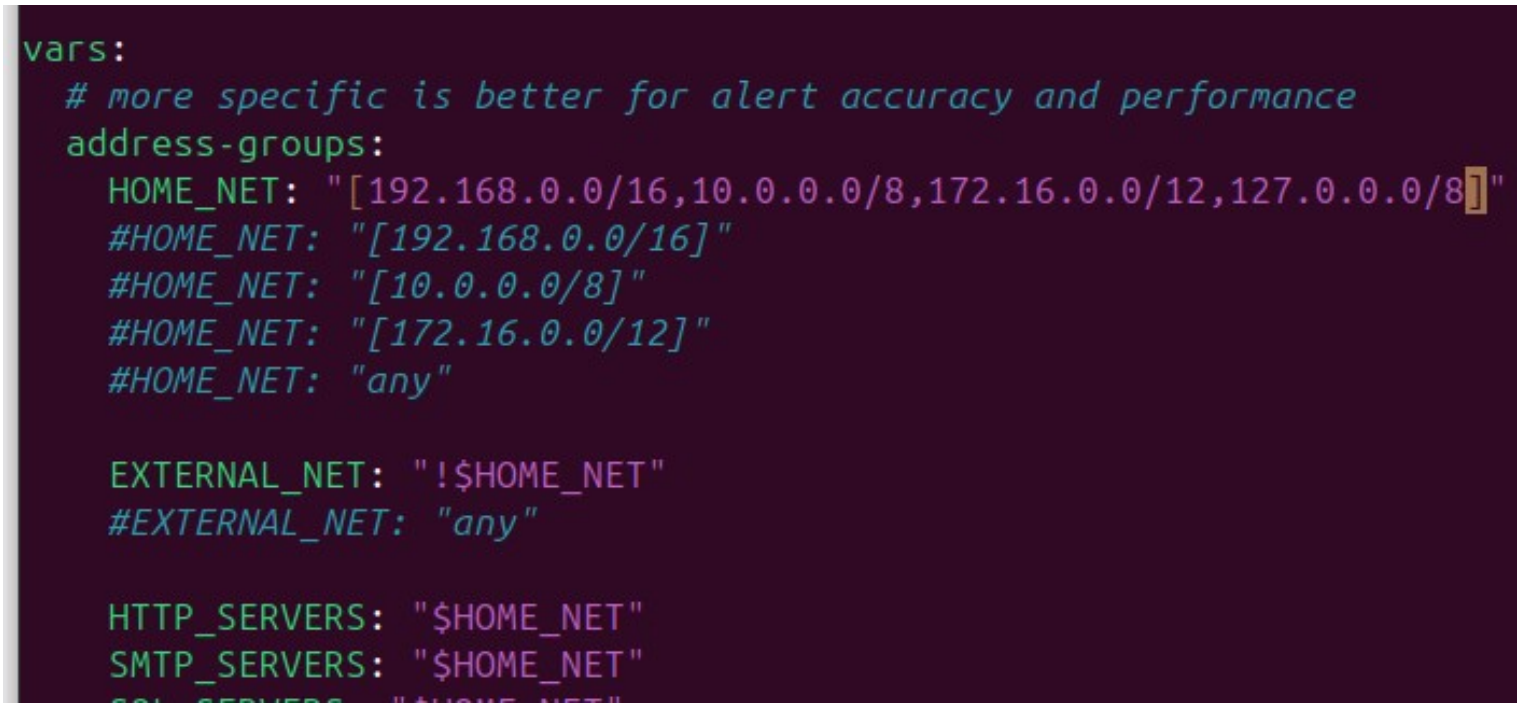
bash

```
sudo nano /etc/suricata/suricata.yaml
```

2.1. Переменная `HOME_NET`

Для корректной работы правил убедитесь, что в секции `vars` → `address-groups` указана ваша локальная сеть (включая loopback для удобства или просто стандартные диапазоны):

yaml



```yaml
vars:
  # more specific is better for alert accuracy and performance
  address-groups:
    HOME_NET: "[192.168.0.0/16,10.0.0.0/8,172.16.0.0/12,127.0.0.0/8]"
    #HOME_NET: "[192.168.0.0/16]"
    #HOME_NET: "[10.0.0.0/8]"
    #HOME_NET: "[172.16.0.0/12]"
    #HOME_NET: "any"

    EXTERNAL_NET: "!$HOME_NET"
    #EXTERNAL_NET: "any"

    HTTP_SERVERS: "$HOME_NET"
    SMTP_SERVERS: "$HOME_NET"
```

2.2. Сетевой интерфейс (af-packet)

Найтисекцию `af-packet`. Чтобы Suricata видела локальный трафик, указан интерфейс **`lo`** (loopback):

yaml

```
af-packet:
  - interface: lo
    cluster-id: 99
    cluster-type: cluster_flow
    defrag: yes
    tpacket-v3: yes
```

```
# Linux high speed capture support
af-packet:
  - interface: lo
    # Number of receive threads. "auto" uses the number of cores
    #threads: auto
    # Default clusterid. AF_PACKET will load balance packets base
    cluster-id: 99
    # Default AF_PACKET cluster type. AF_PACKET can load balance
    # This is only supported for Linux kernel > 3.1
    # possible value are:
    #  * cluster_flow: all packets of a given flow are sent to th
    #  * cluster_cpu: all packets treated in kernel by a CPU are
```

2.3. Подключение `local.rules`

В самом низу файла найдите секцию `rule-files` и добавьте полный путь к вашему файлу правил:

yaml

```
rule-files:
  - suricata.rules
  - /etc/suricata/rules/local.rules
```

```
##
## Configure Suricata to load Suricata-Update managed rules.
##

default-rule-path: /var/lib/suricata/rules

rule-files:
  - suricata.rules
  - /etc/suricata/rules/local.rules
##
## Auxiliary configuration files.
##

classification-file: /etc/suricata/classification.config
reference-config-file: /etc/suricata/reference.config
# threshold-file: /etc/suricata/threshold.config
```

2.4. Проверка конфигурации

Перед запуском проверяем настройки:

```
sudo suricata -T -c /etc/suricata/suricata.yaml -v
```

```
admen@ubuntu:~/Рабочий стол$ sudo suricata -T -c /etc/suricata/suricata.yam
l -v
Notice: suricata: This is Suricata version 8.0.7 RELEASE running in SYSTEM
mode
Info: cpu: CPUs/cores online: 4
Info: suricata: Running suricata under test mode
Info: suricata: Setting engine mode to IDS mode by default
Info: exception-policy: master exception-policy set to: auto
Info: suricata: Preparing unexpected signal handling
Info: logopenfile: fast output device (regular) initialized: fast.log
Info: logopenfile: eve-log output device (regular) initialized: eve.json
Info: logopenfile: stats output device (regular) initialized: stats.log
Info: detect: 2 rule files processed. 53272 rules successfully loaded, 0 ru
les failed, 0 rules skipped
Info: threshold-config: Threshold config parsed: 0 rule(s) found
Info: detect: 53277 signatures processed. 1214 are IP-only rules, 4530 are
inspecting packet payload, 47295 inspect application layer, 110 are decoder
 event only
Notice: suricata: Configuration provided was successfully loaded. Exiting.
admen@ubuntu:~/Рабочий стол$
```

2.5. Запуск службы

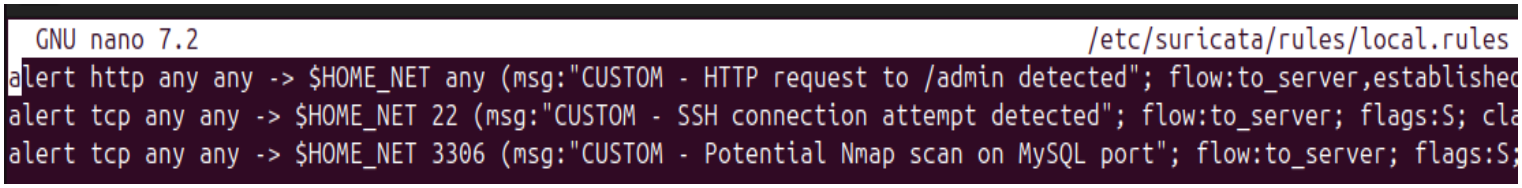
```
sudo systemctl restart suricata
sudo systemctl status suricata
```

```
admen@ubuntu:~/work/suricataHW$ sudo suricata -T -c /etc/suricata/suricata.yaml -v
Notice: suricata: This is Suricata version 8.0.7 RELEASE running in SYSTEM mode
Info: cpu: CPUs/cores online: 4
Info: suricata: Running suricata under test mode
Info: suricata: Setting engine mode to IDS mode by default
Info: exception-policy: master exception-policy set to: auto
Info: suricata: Preparing unexpected signal handling
Info: logopenfile: fast output device (regular) initialized: fast.log
Info: logopenfile: eve-log output device (regular) initialized: eve.json
Info: logopenfile: stats output device (regular) initialized: stats.log
Info: detect: 1 rule files processed. 53269 rules successfully loaded, 0 rules failed, 0 rules skipped
Info: threshold-config: Threshold config parsed: 0 rule(s) found
Info: detect: 53274 signatures processed. 1214 are IP-only rules, 4530 are inspecting packet payload, 47294 inspect application layer, 110 are decoder event only
Notice: suricata: Configuration provided was successfully loaded. Exiting.
admen@ubuntu:~/work/suricataHW$ sudo systemctl restart suricata
sudo systemctl status suricata
● suricata.service - Suricata IDS/IPS/NSM/FW daemon
     Loaded: loaded (/usr/lib/systemd/system/suricata.service; enabled; preset: enabled)
     Active: active (running) since Sun 2026-10-25 18:39:00 UTC; 14ms ago
       Docs: man:suricata(8)
             man:suricatasc(8)
             https://suricata.io/documentation/
    Process: 19108 ExecStartPre=/bin/rm -f /run/suricata.pid (code=exited, status=0/SUCCESS)
   Main PID: 19110 (suricata)
      Tasks: 1 (limit: 14242)
     Memory: 660.0K (peak: 1.6M)
        CPU: 20ms
     CGroup: /system.slice/suricata.service
             └─19110 /usr/bin/suricata --af-packet -c /etc/suricata/suricata.yaml --pidfile /run/suricata.pid --user suricata --group suricata

окт 25 18:39:00 ubuntu systemd[1]: Starting suricata.service - Suricata IDS/IPS/NSM/FW daemon...
окт 25 18:39:00 ubuntu systemd[1]: Started suricata.service - Suricata IDS/IPS/NSM/FW daemon.
admen@ubuntu:~/work/suricataHW$
```

3. Создание и тестирование трёх сигнатур

Все правила добавляются в файл `/etc/suricata/rules/local.rules`.



Подготовка к тестированию:
Откройте отдельный терминал и запустите простой веб-сервер на Python, который будет отвечать на запросы (оставьте его работать):

bash

```
sudo python3 -m http.server 80
```

Правило 1: Обнаружение HTTP-запроса к подозрительному пути (`/admin`)

Правило (SID 1000001):

text

```
alert http any any -> $HOME_NET any (msg:"CUSTOM - HTTP
request to /admin detected"; flow:to_server,established;
content:"GET"; http_method; content:"/admin"; http_uri;
classtype:attempted-recon; sid:1000001; rev:1;)
```

Запуск живого мониторинга лога (Будет висеть в отдельном терминале):
bash

```
sudo tail -f /var/log/suricata/fast.log | grep 1000001
```

Тестирование: В новом окне терминала выполните запрос через `curl` на loopback-адрес:

bash

```
curl http://127.0.0.1/admin
```

*(*Сервер вернет ошибку 404, но главное — запрос уйдет).

Правило 2: Обнаружение SSH-подключения (порт 22)

Правило (SID 1000002):

text

```
alert tcp any any -> $HOME_NET 22 (msg:"CUSTOM - SSH
connection attempt detected"; flow:to_server; flags:S;
classtype:attempted-recon; sid:1000002; rev:1;)
```

Тестирование: Выполните попытку подключения к SSH на loopback:

bash

```
ssh -o ConnectTimeout=2 testuser@127.0.0.1
```

Запуск живого мониторинга лога:

bash

```
sudo tail -f /var/log/suricata/fast.log | grep 1000002
```

Правило 3: Обнаружение сканирования портов (Nmap SYN scan)

Правило (SID 1000003):

text

```
alert tcp any any -> $HOME_NET 3306 (msg:"CUSTOM -
Potential Nmap scan on MySQL port"; flow:to_server;
flags:S; threshold: type limit, track by_src, count 1,
seconds 60; classtype:attempted-recon; sid:1000003;
rev:1;)
```

Тестирование: Выполните сканирование порта с помощью `nmap` (если не установлен: `sudo apt install nmap -y`):

bash

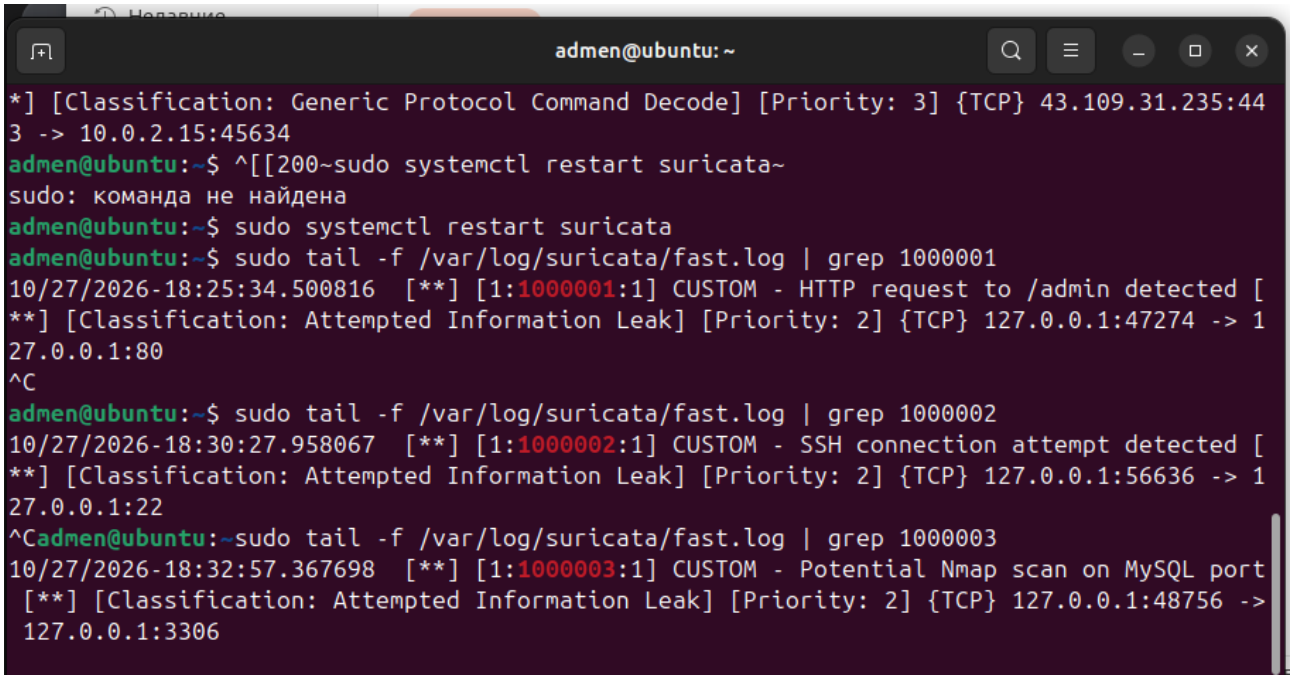
```
nmap -p 3306 127.0.0.1
```

Проверка лога:

bash

```
sudo jq 'select(.alert.signature_id == 1000003)'
/var/log/suricata/eve.json
```

Лог обнаружений и вводимые запросы




```
admen@ubuntu:~$ curl http://127.0.0.1/admin
<!DOCTYPE HTML>
<html lang="en">
    <head>
        <meta charset="utf-8">
        <title>Error response</title>
    </head>
    <body>
        <h1>Error response</h1>
        <p>Error code: 404</p>
        <p>Message: File not found.</p>
        <p>Error code explanation: 404 - Nothing matches the given URI.</p>
    </body>
</html>
admen@ubuntu:~$ ssh -o ConnectTimeout=2 testuser@127.0.0.1
ssh: connect to host 127.0.0.1 port 22: Connection refused
admen@ubuntu:~$ nmap -p 3306 127.0.0.1
Команда «nmap» не найдена, но может быть установлена с помощью:
sudo snap install nmap  # version 7.95, or
sudo apt  install nmap  # version 7.94+git20230807.3be01efb1+dfsg-2
См. 'snap info nmap', чтобы посмотреть дополнительные версии.
admen@ubuntu:~$ sudo snap install nmap  # version 7.95
[sudo] пароль для admen:
nmap 7.992 от Maximiliano Bertacchini (maxiberta😵) installed
admen@ubuntu:~$ nmap -p 3306 127.0.0.1
Starting Nmap 7.992 ( https://nmap.org ) at 2026-10-27 18:32 +0000
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000030s latency).

PORT     STATE  SERVICE
3306/tcp closed mysql

Nmap done: 1 IP address (1 host up) scanned in 0.04 seconds
admen@ubuntu:~$
```